AIPI 510 / MODULE PROJECT 2

Can one minute represent five?

Continuous vs distributed observation at Abbey Road



Abbey Road crossing, context photograph
Source: Abbey Road Studios (2019)

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One window shows how estimates can differ

One observed five-minute window: 10:25–10:30 AM, November 3

43

crossings during the full 5 minutes

35

A estimate

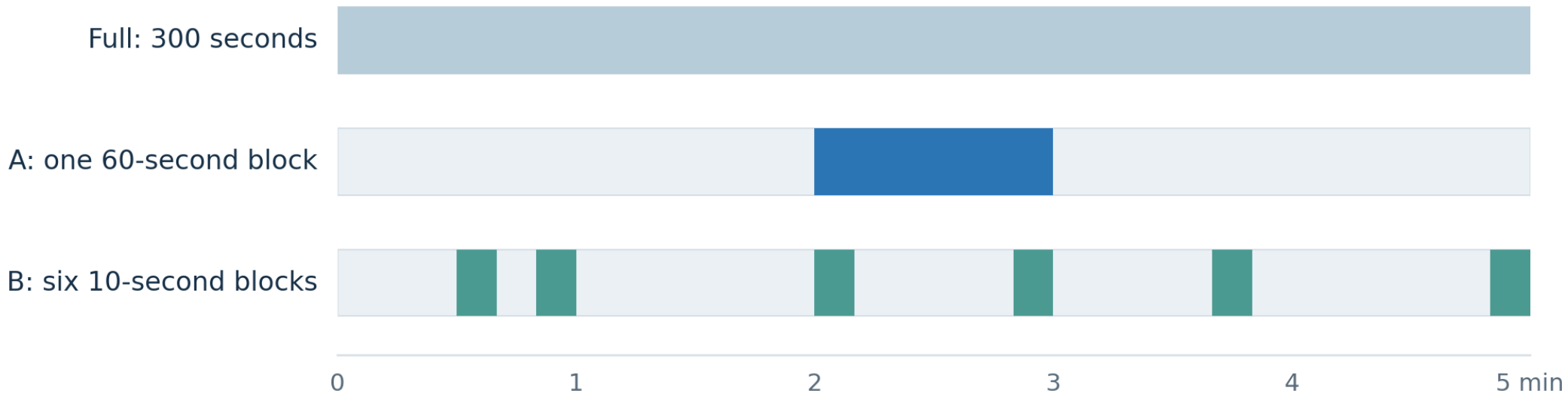
95

B estimate

The sampling intervals can land on very different moments.

Two sampling methods

Both methods use 60 seconds out of a five-minute window.



Estimated total = sample count × 5 • Error = |estimate − full count|

36 windows, sampled before we counted

One camera · one London day · eight hourly strata

36

five-minute windows

828

recorded crossings

60 s

observed by each method

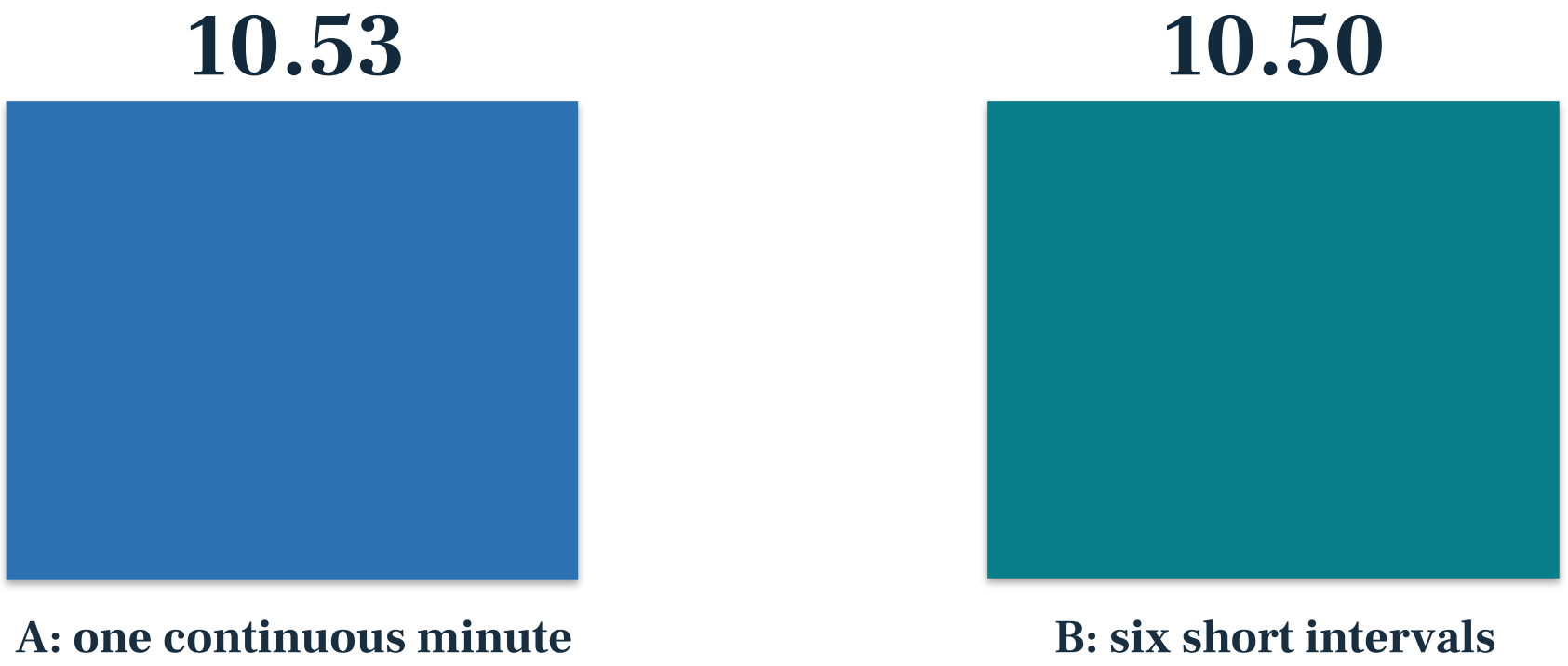
Randomized 5-minute slots and interval positions

Seed 20 261 103 · assumed $d_z = 0.5$ · nominal independent-pair power ≈ 83%

$H_0: \mu_D = 0$ $H_1: \mu_D \neq 0$ D = absolute error A − absolute error B

The average errors were almost identical

Absolute estimation error in pedestrians per five-minute window

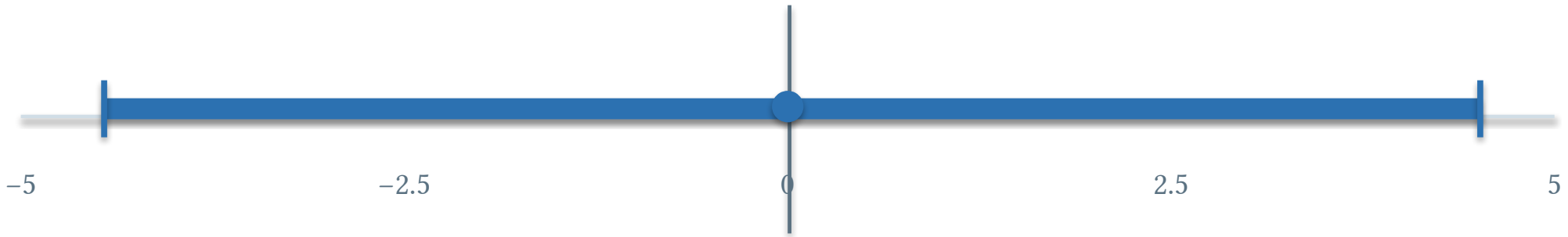


15 windows favor A • 15 favor B • 6 ties

The mean difference remains uncertain

Paired mean difference (A absolute error − B absolute error)

95% confidence interval: −4.46 to +4.51 pedestrians



+0.03	0.990	36
mean difference	paired t-test p-value	paired windows

Cohen's d_z = 0.002 · paired and hour-cluster sign-flip p = 1.000

Limitations

All 36 retained after the observer confirmed visibility during counting.

01 **One camera and one day**

We sampled eight hours on one Tuesday.

02 **Nearby windows may correlate**

There are 36 windows within eight hourly groups.

03 **Varying playback speeds**

Groups and reaction delay can affect timed counts.

04 **One observer and one event log**

No independent recount; reference and samples share annotations.

THE TAKEAWAY

Neither method showed a clear accuracy advantage

Similar averages do not establish equivalent accuracy.

Next: more days, more sites, independent counting checks.

Source: abbeyroad.com/crossing | 36 observer-confirmed windows, Nov 3, 2026